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Privacy-Aware IPv6 Security Monitoring for IoT Devices under Address Rotation

Kavin Krishnan C
kavinkrishnanc.cs24@rvce.edu.in

Team KMK · Krupanka R, Kavin Krishnan C · Mentor: Dr. Manonmani S, Assoc. Prof., Dept. of CSE
RV College of Engineering (RVCE), Bengaluru

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Problem Description

IPv6 address rotation breaks IP-based security monitoring.

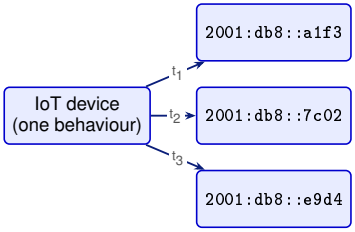
- Devices rotate IPv6 interface identifiers (privacy extensions) and ISPs rotate prefixes, so a single device can appear under many addresses over time.
- Snoopers and compromised IoT devices exploit this: the identity seen by the network is assumed to change every rotation cycle.
- On a **static** network, anomalies are easy to spot from bandwidth or throughput per address. On a **dynamic** one, per-address logs, blocklists and rate limits fragment and lose meaning.
- Monitoring must still respect user privacy: no payload inspection, no permanent tracking of innocent users.

Our approach in one line

Stop keying identity on the IPv6 address. Learn each device's **behaviour profile** with AI and anchor the address-to-profile history on a **blockchain ledger**, so a device stays trackable across rotations.

Background

- IPv6 hosts self-configure addresses (SLAAC). **Temporary addresses** with randomised interface identifiers are generated to stop long-term tracking; by default they are preferred for about a day and valid for about two days [1].
- Stable-but-opaque identifiers per prefix [2] and ISP prefix changes add further churn.
- Result: the address is no longer a reliable identity.
- **IoT devices are different from laptops and phones.** They do a few fixed tasks, so their traffic (sizes, timing, destinations) is repetitive and distinctive [3].



Same device, three addresses:
an IP-keyed log sees three strangers.

Literature & RFC Review

Source	What it gives us	Gap for this problem
RFC 8981 [1] Temporary addresses	Defines address rotation and its lifetimes	Rotation defeats address-keyed monitoring
RFC 7217 [2] Stable opaque IIDs	Per-prefix stable identifiers	Still changes when prefix changes
Sivanathan et al. [3]	IoT devices can be classified from traffic characteristics	Classifies device *types*, not tracking across rotations
Rye et al. [4]	Shows IPv6 prefix rotation can be defeated for tracking	Active measurement, not a privacy-aware defender-side design
Nakamoto [5], Fabric [6]	Tamper-evident, append-only shared ledger; permissioned chains	Not applied to identity linkage of rotating addresses

Gap: no single design maps rotating addresses to a behavioural profile, keeps an auditable history of that mapping, and stays metadata-only.

Proposed Solution

Map behaviour, not the address. A four-stage pipeline:

1. **Capture & features.** Open-source tools (e.g. tcpdump, Zeek) measure per-flow bandwidth and throughput in time windows. Only metadata is kept: packet size, inter-arrival time, up/down ratio, protocol and port mix, duty cycle. No payloads.
2. **AI behaviour model.** A model trained dynamically on these windows turns each device's activity into a behaviour fingerprint (embedding).
3. **Identity resolution.** When a new address appears, its first windows are compared with known profiles. A close match links the address to that profile; otherwise a new profile is created. Profile IDs are pseudonymous hashes.
4. **Blockchain ledger.** Each link (profile ID, address, time, fingerprint digest) is appended to a permissioned ledger. History is tamper-evident, so an investigator can replay what one profile did across every rotation.

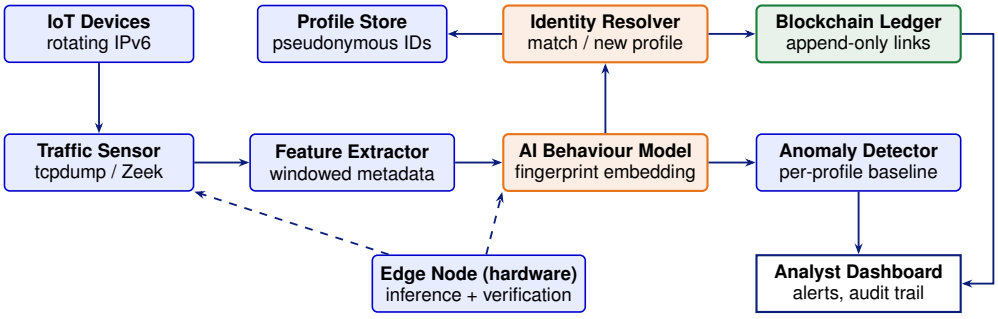
Alerts fire when a profile drifts from its own baseline, whichever address it uses. The logic is also targeted at an edge (hardware) node.

Optimization Proposed

Aspect	Before (IP-keyed)	After (behaviour + ledger)
Identity key	IPv6 address	Behavioural profile ID
Tracking across rotation	Lost at every cycle	Continuous via fingerprint matching
Detection baseline	Static throughput thresholds	Adaptive per-profile baselines
Evidence integrity	Mutable local logs	Append-only, tamper-evident ledger
User privacy	Raw IPs and traffic logs	Metadata only, pseudonymous hashed IDs
Compute placement	Central collector	Lightweight inference at the edge

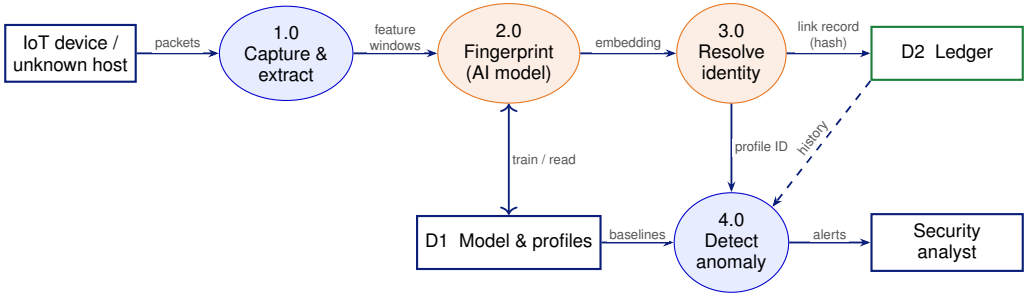
Evaluation metrics we will report: re-identification accuracy across rotations, false-alarm rate, detection latency, and ledger write overhead.

Solution Architecture & Design



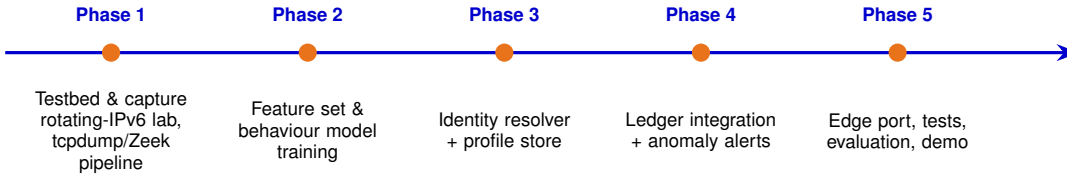
Orange = AI / identity logic · green = ledger · dashed = edge deployment of the same modules

Data Flow Diagram



Only hashed profile IDs and feature digests reach the ledger. Raw packets never leave process 1.0.

Deployment Timeline



- **Method:** iterate in short cycles. Each phase ends with a runnable demo on the lab testbed.
- **Roles:** Kavin: hardware implementation, verification, testing and simulations. Krupanka: software application, AI integration, algorithms in hardware.
- **Exit criteria:** the same device is re-linked after forced address rotation; the ledger history verifies; alerts fire on injected anomalies.

Bibliography

[1] F. Gont, S. Krishnan, T. Narten, R. Draves. *Temporary Address Extensions for Stateless Address Autoconfiguration in IPv6*. RFC 8981, IETF, 2021.

[2] F. Gont. *A Method for Generating Semantically Opaque Interface Identifiers with IPv6 SLAAC*. RFC 7217, IETF, 2014.

[3] A. Sivanathan et al. “Classifying IoT Devices in Smart Environments Using Network Traffic Characteristics.” *IEEE Transactions on Mobile Computing*, 18(8), 2019.

[4] E. Rye, R. Beverly, kc claffy. “Follow the Scent: Defeating IPv6 Prefix Rotation Privacy.” *ACM IMC*, 2021.

[5] S. Nakamoto. *Bitcoin: A Peer-to-Peer Electronic Cash System*. 2008.

[6] E. Androulaki et al. “Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains.” *EuroSys*, 2018.

Thank You for Your Attention!

Kavin Krishnan C · kavinkrishnanc.cs24@rvce.edu.in
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